lemma lem:x0-geometry

statement

One has

$$(\operatorname{Im} X_0)^\perp = \ker X_0.$$

Hence

$$\dim(V_0/\operatorname{Im} X_0) = \dim\ker X_0 = c,$$

the quotient map identifies the fixed complement V_+ with $V_0/\operatorname{Im} X_0$, and the pairing

$$V_+ \times \ker X_0 \to F, \qquad (v,w) \mapsto \langle v,w\rangle_0$$

is perfect. In particular, for every isomorphism $S : L_1' \xrightarrow{\sim} V_+$, the restriction

$$S^\vee|_{\ker X_0} : \ker X_0 \xrightarrow{\sim} L_1$$

is an isomorphism.

proof

If $w \in \ker X_0$ and $y = X_0 z \in \operatorname{Im} X_0$, then

$$\langle w, y\rangle_0 = \langle w, X_0 z\rangle_0 = -\langle X_0 w, z\rangle_0 = 0,$$

so $w \in (\operatorname{Im} X_0)^\perp$. Therefore

$$\ker X_0 \subseteq (\operatorname{Im} X_0)^\perp.$$

Now

$$\dim(\operatorname{Im} X_0)^\perp = \dim V_0 - \dim\operatorname{Im} X_0 = \dim\ker X_0,$$

so the inclusion above is an equality:

$$(\operatorname{Im} X_0)^\perp = \ker X_0.$$

Since $V_0 = V_+ \oplus \operatorname{Im} X_0$, the quotient map $V_0 \to V_0/\operatorname{Im} X_0$ identifies V_+ with $V_0/\operatorname{Im} X_0$, hence $\dim V_+ = c$. The equality $(\operatorname{Im} X_0)^\perp = \ker X_0$ says exactly that the induced pairing

$$(V_0/\operatorname{Im} X_0) \times \ker X_0 \to F$$

is non-degenerate, so the pairing $V_+ \times \ker X_0 \to F$ is perfect.

Finally, by definition,

$$\lambda(S^\vee v, a') = -\langle v, Sa'\rangle_0.$$

If $v \in \ker X_0$ and $S^\vee v = 0$, then $\langle v, Sa'\rangle_0 = 0$ for every $a' \in L_1'$. Since $S(L_1') = V_+$ and the pairing $V_+ \times \ker X_0$ is perfect, this forces $v = 0$. Because both spaces have dimension c, $S^\vee|_{\ker X_0}$ is an isomorphism.

lemma lem:parametrize-x0-plus-u

statement

For $C \in \mathrm{Hom}_F(E', V_0)$ let $C^\vee : V_0 \to E$ be defined by

$$\lambda(C^\vee v, e') = -\langle v, Ce'\rangle_0.$$

For $B \in \mathrm{Hom}_F(E', E)$ satisfying

$$\lambda(Bu, v) + \epsilon\lambda(Bv, u) = 0 \qquad (u, v \in E'),$$

define

$$X_{C,B}(e + v + e') := C^\vee v + Be' + X_0 v + Ce'.$$

Then $X_{C,B} \in X_0 + \mathfrak{u}$.

Conversely, every element of $X_0 + \mathfrak{u}$ is uniquely of the form $X_{C,B}$.

In particular, for the special choice

$$C|_{L_1'} = S, \qquad C|_{L_0'} = 0, \qquad B|_{L_1'} = 0, \qquad B|_{L_0'} = T,$$

one gets exactly the operator $X_{S,T}$ from the problem statement.

proof

Relative to the decomposition $V = E \oplus V_0 \oplus E'$, an element $Y \in \mathfrak{u}$ has the shape

$$Y = \begin{pmatrix} 0 & A & B \\ 0 & 0 & C \\ 0 & 0 & 0 \end{pmatrix}$$

with $A : V_0 \to E$, $B : E' \to E$, $C : E' \to V_0$. The condition $Y \in \mathfrak{g}$ is equivalent to skew-adjointness:

$$\langle Yx, y\rangle + \langle x, Yy\rangle = 0.$$

Testing this on $(v, e') \in V_0 \times E'$ gives

$$\lambda(Av, e') + \langle v, Ce'\rangle_0 = 0,$$

so $A = C^\vee$. Testing it on $(u, v) \in E' \times E'$ gives

$$\lambda(Bu, v) + \epsilon\lambda(Bv, u) = 0.$$

Thus every $Y \in \mathfrak{u}$ is uniquely $Y_{C,B}$ with

$$Y_{C,B}(e + v + e') = C^\vee v + Be' + Ce'.$$

Hence every element of $X_0 + \mathfrak{u}$ is uniquely

$$X_0 + Y_{C,B} = X_{C,B}.$$

Now $X_{C,B} - X_0 = Y_{C,B}$ certainly annihilates E, maps $E^\perp = E \oplus V_0$ into E, and maps V into $E^\perp$, so $Y_{C,B} \in \mathfrak{u}$. Therefore $X_{C,B} \in X_0 + \mathfrak{u}$.

Taking C and B as in the statement gives exactly the formula

$$X_{S,T}(e + v + a' + b') = (S^\vee v + Tb') + (X_0 v + Sa')$$

from the problem statement. This proves assertion (1) of the problem.

lemma lem:unipotent-conjugation

statement

For $D \in \mathrm{Hom}_F(E', V_0)$ define

$$u_D(e + v + e') := \left(e + D^\vee v + \frac{1}{2} D^\vee D\, e'\right) + (v + De') + e'.$$

Then $u_D \in P$. Moreover, for every pair (C, B) as above,

$$u_D\, X_{C,B}\, u_D^{-1} = X_{C - X_0 D,\ B + D^\vee C - C^\vee D - D^\vee X_0 D}.$$

proof

In block form,

$$u_D = \begin{pmatrix} 1 & D^\vee & \frac{1}{2} D^\vee D \\ 0 & 1 & D \\ 0 & 0 & 1 \end{pmatrix}.$$

To check that u_D preserves the form, it is enough to test it on pairs involving the E'-summand. For $u, v \in E'$ one gets

$$\lambda\left(\frac{1}{2} D^\vee D\, u, v\right) + \epsilon\lambda\left(\frac{1}{2} D^\vee D\, v, u\right) + \langle Du, Dv\rangle_0 = -\frac{1}{2}\langle Du, Dv\rangle_0 - \frac{\epsilon}{2}\langle Dv, Du\rangle_0 + \langle Du, Dv\rangle_0 = 0,$$

because $\langle Dv, Du\rangle_0 = \epsilon\langle Du, Dv\rangle_0$. The mixed (V_0, E')-terms cancel by the defining property of $D^\vee$. Hence $u_D \in P$.

Also

$$X_{C,B} = \begin{pmatrix} 0 & C^\vee & B \\ 0 & X_0 & C \\ 0 & 0 & 0 \end{pmatrix}.$$

Multiplying the two block matrices and simplifying gives

$$u_D\, X_{C,B}\, u_D^{-1} = \begin{pmatrix} 0 & (C - X_0 D)^\vee & B + D^\vee C - C^\vee D - D^\vee X_0 D \\ 0 & X_0 & C - X_0 D \\ 0 & 0 & 0 \end{pmatrix},$$

which is the stated formula.

proposition prop:p-normal-form

statement

Let $x = X_{C,B} \in X_0 + \mathfrak{u}$, and let

$$\overline{C} : E' \to V_0 / \operatorname{Im} X_0 \simeq V_+$$

be the composition of C with the quotient map.

Assume that $\operatorname{rank} \overline{C} = c$. Then x is P-conjugate to some $X_{S,T}$ with $T \in \mathscr{T}$.

More precisely:

1. After P-conjugation one may arrange

$$C|_{L_1'} = S, \qquad C|_{L_0'} = 0.$$

2. Keeping this C fixed, one may further P-conjugate so that all B-blocks involving L_1' vanish, and the remaining $L_0' \to L_0$ block is some $T \in \mathscr{T}$.
3. The residual action of the Levi factor on the $L_0 \oplus L_0'$-block is

$$T \longmapsto (h^{-1})^\vee T h \qquad (h \in \mathrm{GL}(L_0')),$$

   so the resulting P-orbit is determined exactly by the isometry class of the bilinear space (L_0', B_T).

proof

Because $\operatorname{rank} \overline{C} = c = \dim(V_0 / \operatorname{Im} X_0)$, the map $\overline{C}$ is surjective and $\ker \overline{C}$ has dimension l. By a Levi element of P acting through $\mathrm{GL}(E')$, we may assume

$$\ker \overline{C} = L_0'.$$

Then $\overline{C}|_{L_1'} : L_1' \to V_0 / \operatorname{Im} X_0 \simeq V_+$ is an isomorphism. Using another Levi element on L_1', we may assume that this quotient map is exactly the fixed isomorphism induced by $S : L_1' \to V_+$.

Now $C - S$ takes values in $\operatorname{Im} X_0$, so there exists $D : E' \to V_0$ with

$$X_0 D = C - S.$$

By Lemma `lem:unipotent-conjugation`, conjugation by u_D replaces C by

$$C - X_0 D = S,$$

that is,

$$C|_{L_1'} = S, \qquad C|_{L_0'} = 0.$$

We now keep this C fixed and write B in blocks:

$$B = \begin{pmatrix} B_{11} & B_{10} \\ B_{01} & B_{00} \end{pmatrix} : L_1' \oplus L_0' \to L_1 \oplus L_0.$$

Take first $D_0 : L_0' \to \ker X_0$. Because the pairing $V_+ \times \ker X_0$ is perfect, the map

$$D_0 \longmapsto (D_0^\vee S) : L_1' \to L_0$$

is an isomorphism

$$\operatorname{Hom}(L_0', \ker X_0) \xrightarrow{\sim} \operatorname{Hom}(L_1', L_0).$$

Hence we may choose D_0 so that the $L_1' \to L_0$ block of $B + D_0^\vee C - C^\vee D_0$ is zero. After that, the skew relation

$$\lambda(Bu, v) + \epsilon\lambda(Bv, u) = 0$$

forces the opposite block $B_{10} : L_0' \to L_1$ to vanish as well.

Next take $D_1 : L_1' \to \ker X_0$. Let

$$b_{11}(u, v) := \lambda(B_{11}u, v).$$

Then $b_{11}(v, u) = -\epsilon b_{11}(u, v)$. Again by the perfect pairing $V_+ \times \ker X_0$, there is a unique D_1 such that

$$\langle Su, D_1 v\rangle_0 = \frac{1}{2}\, b_{11}(u, v) \qquad (u, v \in L_1').$$

For this choice one computes

$$\lambda\big((B_{11} + D_1^\vee S - S^\vee D_1)u, v\big) = b_{11}(u, v) - \frac{1}{2} b_{11}(u, v) + \frac{\epsilon}{2} b_{11}(v, u) = 0,$$

so the new $L_1' \to L_1$ block also vanishes.

Therefore, after these conjugations, only the $L_0' \to L_0$ block survives. Call it T. The skew relation on B becomes exactly

$$\lambda(Tu, v) + \epsilon\lambda(Tv, u) = 0,$$

so $T \in \mathscr{T}$, and the resulting element is precisely $X_{S,T}$.

Finally, if $h \in \mathrm{GL}(L_0')$ and the Levi element acts on L_0 by $(h^{-1})^\vee$, then it sends T to $(h^{-1})^\vee Th$. This is exactly the change of coordinates on the bilinear form

$$B_T(u, v) = \lambda(Tu, v),$$

so isometric residual forms give P-conjugate normalized representatives.

Conversely, suppose

$$pX_{S,T_1}p^{-1} = X_{S,T_2}$$

for some $p \in P$. Write $p = u_D m$ with u_D as in Lemma `lem:unipotent-conjugation` and m in the Levi factor of P. Write m on E' as $d \in \mathrm{GL}(E')$, on V_0 as $g_0 \in G_0$, and on E as $(d^{-1})^\vee$. A direct block calculation gives

$$mX_{C,B}m^{-1} = X_{g_0Cd^{-1},\,(d^{-1})^\vee Bd^{-1}}.$$

For the normalized C-block

$$C_0|_{L_1'} = S, \qquad C_0|_{L_0'} = 0,$$

the quotient class $\overline{C_0} : E' \to V_0/\operatorname{Im} X_0$ has kernel exactly L_0'. Since unipotent conjugation changes C by

$$C \longmapsto C - X_0 D,$$

it does not change $\overline{C}$. Hence m must preserve $\ker \overline{C_0} = L_0'$; write

$$h := d|_{L_0'} \in \mathrm{GL}(L_0').$$

Let

$$C' := g_0 C_0 d^{-1}.$$

Since both C' and the final normalized block C_0 have kernel L_0', we have

$$C'|_{L_0'} = 0, \qquad (C' - C_0)|_{L_0'} = 0.$$

But $C' - C_0 = X_0 D$, so

$$X_0 D(L_0') = 0,$$

hence $D(L_0') \subseteq \ker X_0$. Lemma `lem:unipotent-conjugation` shows that u_D changes B by

$$B \longmapsto B + D^\vee C' - C'^\vee D - D^\vee X_0 D.$$

When restricted to the source L_0', the first term $D^\vee C'$ is zero because $C'|_{L_0'} = 0$, and the last term $D^\vee X_0 D$ is zero because $X_0 D(L_0') = 0$. For the middle term, if $w \in \ker X_0$, then $C'^\vee w \in L_1$ because C' vanishes on L_0', so $C'^\vee D(L_0') \subseteq L_1$. Therefore the whole correction term has zero $L_0' \to L_0$ block. Thus the unipotent factor does not change the residual $L_0' \to L_0$ block at all.

It follows that for a normalized representative, the only effect on the residual block T comes from the Levi factor, namely

$$T \longmapsto (h^{-1})^\vee T h.$$

Since $pX_{S,T_1}p^{-1} = X_{S,T_2}$ and both sides are normalized, we get

$$T_2 = (h^{-1})^\vee T_1 h.$$

Equivalently, (L_0', B_{T_1}) and (L_0', B_{T_2}) are isometric. Thus two normalized representatives are P-conjugate if and only if their residual bilinear spaces are isometric.

proposition prop:kernel-image

statement

For every S and every $T \in \mathscr{T}$ one has

$$\ker X_{S,T} = E \oplus \ker T,$$

and

$$\operatorname{Im} X_{S,T} = (L_1 \oplus \operatorname{Im} T) \oplus V_0.$$

Hence

$$\dim \ker X_{S,T} = r + (l - \operatorname{rank} T).$$

proof

Take

$$x = e + v + a' + b' \in E \oplus V_0 \oplus L_1' \oplus L_0'.$$

Then

$$X_{S,T} x = (S^\vee v + Tb') + (X_0 v + Sa').$$

If $X_{S,T} x = 0$, then the V_0-component gives

$$X_0 v + Sa' = 0.$$

Since $X_0 v \in \operatorname{Im} X_0$ and $Sa' \in V_+$, and $V_0 = V_+ \oplus \operatorname{Im} X_0$, both terms must vanish:

$$a' = 0, \qquad v \in \ker X_0.$$

Then the E-component becomes

$$S^\vee v + Tb' = 0.$$

Now $S^\vee v \in L_1$ and $Tb' \in L_0$, so again each term is zero. By Lemma `lem:x0-geometry`, $S^\vee|_{\ker X_0}$ is injective, hence $v = 0$, and therefore $b' \in \ker T$.

So indeed

$$\ker X_{S,T} = E \oplus \ker T.$$

For the image, we first prove the reverse inclusion

$$(L_1 \oplus \operatorname{Im} T) \oplus V_0 \subseteq \operatorname{Im} X_{S,T}.$$

Indeed, if $y \in V_+$, then $y = Sa'$ for some $a' \in L_1'$, and

$$X_{S,T}(a') = y.$$

If $y \in \operatorname{Im} X_0$, choose $v \in V_0$ with $X_0 v = y$. By Lemma `lem:x0-geometry`, the map

$$S^\vee|_{\ker X_0} : \ker X_0 \xrightarrow{\sim} L_1$$

is surjective, so there exists $k \in \ker X_0$ with

$$S^\vee k = S^\vee v.$$

Then

$$X_{S,T}(v - k) = \big(S^\vee v - S^\vee k\big) + \big(X_0 v - X_0 k\big) = y.$$

Since $V_0 = V_+ \oplus \operatorname{Im} X_0$, this proves

$$V_0 \subseteq \operatorname{Im} X_{S,T}.$$

Also, if $e_1 \in L_1$, choose $k \in \ker X_0$ with $S^\vee k = e_1$; then

$$X_{S,T}(k) = e_1.$$

Thus $L_1 \subseteq \operatorname{Im} X_{S,T}$, and clearly

$$X_{S,T}(L_0') = \operatorname{Im} T,$$

so $\operatorname{Im} T \subseteq \operatorname{Im} X_{S,T}$ as well.

Conversely, every value of $X_{S,T}$ has the form

$$\big(S^\vee v + Tb'\big) + \big(X_0 v + Sa'\big),$$

whose E-component lies in $L_1 \oplus \operatorname{Im} T$ and whose V_0-component lies in V_0. Therefore

$$\operatorname{Im} X_{S,T} \subseteq (L_1 \oplus \operatorname{Im} T) \oplus V_0.$$

Hence

$$\operatorname{Im} X_{S,T} = (L_1 \oplus \operatorname{Im} T) \oplus V_0.$$

The dimension formula follows immediately.

lemma lem:local-form-classes

statement

The maximal-rank forms occurring in $\mathscr{T}^\circ$ have only finitely many isometry classes over the allowed fields, and each isometry class is open in $\mathscr{T}^\circ$.

More precisely:

1. If $\epsilon = +1$, then B_T is alternating of maximal rank, and its isometry class is unique.
2. If $\epsilon = -1$ and $F = \mathbb{R}$, then B_T is a non-degenerate symmetric form, classified by its signature; therefore there are finitely many isometry classes, and each class is open in the non-degenerate locus.
3. If $\epsilon = -1$ and F is non-archimedean of characteristic 0, then B_T is a non-degenerate symmetric form, classified by the determinant modulo $F^{\times 2}$ and the Hasse invariant; therefore there are finitely many isometry classes, and each class is open in the non-degenerate locus.

proof

If $\epsilon = +1$, then B_T is alternating. On an even-dimensional space there is only one non-degenerate alternating form up to isometry, and on an odd-dimensional space there is only one alternating form of rank $l-1$ up to isometry: in a suitable basis it is the standard symplectic form on a $2m$-dimensional subspace and 0 on the radical line.

Now suppose $\epsilon = -1$, so B_T is symmetric and non-degenerate on L_0'.

If $F = \mathbb{R}$, Sylvester's law of inertia gives the classification by signature (p, q) with $p + q = l$. There are only finitely many such signatures, and the signature is unchanged under sufficiently small perturbations inside the non-degenerate locus, so each class is open.

If F is non-archimedean of characteristic 0, the standard local classification of non-degenerate quadratic forms gives: a symmetric form of dimension l is determined up to isometry by its determinant in $F^\times / F^{\times 2}$ and its Hasse invariant. Both invariants take only finitely many values, and both are locally constant on the non-degenerate locus. Hence there are finitely many isometry classes, each open.

proposition prop:induced-orbits

statement

The induced G-orbits in $\mathrm{Ind}_P^G(\mathcal{O}_0)$ are exactly the orbits

$$G \cdot X_{S,T} \qquad (T \in \mathscr{T}^\circ),$$

for any fixed choice of isomorphism $S : L_1' \xrightarrow{\sim} V_+$.

proof

Embed G_0 into the Levi factor of P by

$$\iota(g_0)(e + v + e') := e + g_0 v + e'.$$

Then $\iota(g_0) \in P$, and

$$\iota(g_0) X_0 \iota(g_0)^{-1} = g_0 X_0 g_0^{-1}.$$

Also $\mathfrak{u}$ is P-stable: if $p \in P$, then $p(E) = E$ and $p(E^\perp) = E^\perp$, so the defining conditions

$$Y(E) = 0, \qquad Y(E^\perp) \subseteq E, \qquad Y(V) \subseteq E^\perp$$

are preserved by $\mathrm{Ad}(p)$. Therefore

$$\mathcal{O}_0 + \mathfrak{u} = \bigcup_{g_0 \in G_0} \left(\iota(g_0) X_0 \iota(g_0)^{-1} + \mathfrak{u} \right) = \iota(G_0) \cdot (X_0 + \mathfrak{u}),$$

and hence

$$G \cdot (\mathcal{O}_0 + \mathfrak{u}) = G \cdot (X_0 + \mathfrak{u}).$$

So

$$\mathrm{Ind}_P^G(\mathcal{O}_0) = \overline{G \cdot (X_0 + \mathfrak{u})}.$$

We next isolate the open dense part of the slice $X_0 + \mathfrak{u}$. Via Lemma `lem:parametrize-x0-plus-u`, write points of the slice uniquely as

$$x = X_{C,B}.$$

Let

$$U := \{X_{C,B} \in X_0 + \mathfrak{u} : \mathrm{rank}\, \overline{C} = c\}.$$

Since $\overline{C} : E' \to V_0 / \mathrm{Im}\, X_0$ has target dimension c, the condition

$$\mathrm{rank}\, \overline{C} = c$$

is open and dense, so U is open dense in $X_0 + \mathfrak{u}$.

Define

$$\Omega := \bigcup_{T \in \mathscr{T}^\circ} P \cdot X_{S,T} \subseteq X_0 + \mathfrak{u}.$$

We claim that Ω is open dense in $X_0 + \mathfrak{u}$.

To prove openness, cover U by finitely many standard minor charts. Fix a basis of E' adapted to

$$E' = L_1' \oplus L_0'.$$

For each c-element subset I of that basis, let $U_I \subseteq U$ be the locus where the images under $\overline{C}$ of the basis vectors indexed by I form a basis of

$$V_0 / \operatorname{Im} X_0 \simeq V_+.$$

Then

$$U = \bigcup_I U_I,$$

with finitely many I, and each U_I is open.

On U_I, the normalization procedure of Proposition `prop:p-normal-form` can be carried out with a Levi element depending rationally on C: first send the span of the basis vectors indexed by I to L_1', then use the quotient map to identify it with the fixed isomorphism

$$S : L_1' \xrightarrow{\sim} V_+,$$

and finally apply the explicit unipotent corrections from Proposition `prop:p-normal-form`. The output is a rational map

$$T_I : U_I \to \mathscr{T}$$

such that every $x \in U_I$ is P-conjugate to

$$X_{S,T_I(x)}.$$

By item (3) of Proposition `prop:p-normal-form`, two points of U_I lie in the same P-orbit exactly when their residual forms are isometric; in particular,

$$\Omega \cap U_I = \{x \in U_I : \operatorname{rank} T_I(x) \text{ is maximal}\}.$$

Since $\mathscr{T}^\circ$ is open in $\mathscr{T}$, each $\Omega \cap U_I$ is open in U_I. Because the cover by the U_I is finite, Ω is open in U, hence open in $X_0 + \mathfrak{u}$.

To prove density, let $x \in U$. By Proposition `prop:p-normal-form`, x is P-conjugate to some $X_{S,T}$ with $T \in \mathscr{T}$. Since $\mathscr{T}^\circ$ is dense in $\mathscr{T}$, choose a net (or sequence) $T_\nu \in \mathscr{T}^\circ$ with

$$T_\nu \to T.$$

Then

$$X_{S,T_\nu} \to X_{S,T},$$

and after conjugating by the fixed element of P carrying $X_{S,T}$ to x, we get points of Ω converging to x. Thus Ω is dense in U, and since U is dense in $X_0 + \mathfrak{u}$, Ω is dense in the whole slice.

Consequently

$$G \cdot \Omega = \bigcup_{g \in G} g\Omega$$

is open dense in

$$G \cdot (X_0 + \mathfrak{u}) = \bigcup_{g \in G} g(X_0 + \mathfrak{u}),$$

because each translate $g\Omega$ is open dense in $g(X_0 + \mathfrak{u})$. Therefore

$$\mathrm{Ind}_P^G(\mathcal{O}_0) = \overline{G \cdot (X_0 + \mathfrak{u})} = \overline{G \cdot \Omega}.$$

By Lemma `lem:local-form-classes`, there are only finitely many isometry classes of maximal-rank forms on L_0'. Choose representatives

$$T_1, \ldots, T_N \in \mathscr{T}^\circ$$

for these classes, and set

$$\Omega_i := P \cdot X_{S,T_i} \qquad (1 \le i \le N).$$

Now item (3) of Proposition `prop:p-normal-form` gives the P-orbit decomposition

$$\Omega = \bigsqcup_{i=1}^{N} \Omega_i.$$

Let

$$\mathcal{O}_1, \ldots, \mathcal{O}_M$$

be the distinct G-orbits among the finitely many sets

$$G \cdot X_{S,T_i}.$$

Then

$$G \cdot \Omega = \bigcup_{j=1}^{M} \mathcal{O}_j.$$

For each j, the intersection

$$\mathcal{O}_j \cap \Omega$$

is a union of some of the open subsets Ω_i, hence is open in Ω. Moreover

$$\mathcal{O}_j = G \cdot (\mathcal{O}_j \cap \Omega) = \bigcup_{g \in G} g(\mathcal{O}_j \cap \Omega),$$

so each $\mathcal{O}_j$ is open in $G \cdot \Omega$. Since only finitely many $\mathcal{O}_j$ occur and they partition $G \cdot \Omega$, each $\mathcal{O}_j$ is also closed in $G \cdot \Omega$.

Because

$$\mathrm{Ind}_P^G(\mathcal{O}_0) \;=\; \overline{G \cdot \Omega} = \overline{\bigcup_{j=1}^{M} \mathcal{O}_j} \subseteq \bigcup_{j=1}^{M} \overline{\mathcal{O}_j},$$

every G-orbit in the induced set lies in the closure of at least one $\mathcal{O}_j$.

It remains to prove that the $\mathcal{O}_j$ are exactly the maximal orbits. First, any orbit

$$\mathcal{O} \subseteq \mathrm{Ind}_P^G(\mathcal{O}_0)$$

which is not one of the $\mathcal{O}_j$ lies in some $\overline{\mathcal{O}_j}$, hence is strictly below $\mathcal{O}_j$ in the closure order and therefore is not maximal.

Conversely, let $\mathcal{O}_j$ be fixed and suppose

$$\mathcal{O}_j \subseteq \overline{\mathcal{O}}$$

for some G-orbit $\mathcal{O}$ in the induced set. As above, $\mathcal{O}$ lies in $\overline{\mathcal{O}_k}$ for some k, so

$$\mathcal{O}_j \subseteq \overline{\mathcal{O}_k}.$$

Since $\mathcal{O}_k$ is closed in $G \cdot \Omega$, this forces

$$\mathcal{O}_j \subseteq \overline{\mathcal{O}_k} \cap G \cdot \Omega = \mathcal{O}_k.$$

But the $\mathcal{O}_j$ are disjoint G-orbits, so necessarily

$$\mathcal{O}_j = \mathcal{O}_k.$$

Hence $\mathcal{O} \subseteq \overline{\mathcal{O}_j}$ and also $\mathcal{O}_j \subseteq \overline{\mathcal{O}}$. Since an orbit is the unique open dense orbit in its own closure, this implies

$$\mathcal{O} = \mathcal{O}_j.$$

Therefore each $\mathcal{O}_j$ is maximal.

Thus the induced G-orbits are exactly the orbits

$$G \cdot X_{S,T} \qquad (T \in \mathscr{T}^\circ).$$

proposition prop:s-independence-and-orbit-criterion

statement

Fix $T \in \mathscr{T}$. For any two isomorphisms $S_1, S_2 : L_1' \xrightarrow{\sim} V_+$ one has

$$G \cdot X_{S_1,T} = G \cdot X_{S_2,T}.$$

More generally, for $T_1, T_2 \in \mathscr{T}^\circ$ and $S_1, S_2 : L_1' \xrightarrow{\sim} V_+$,

$$G \cdot X_{S_1,T_1} = G \cdot X_{S_2,T_2}$$

if and only if the bilinear spaces (L_0', B_{T_1}) and (L_0', B_{T_2}) are isometric.

proof

Choose $h_1 := S_2^{-1} S_1 \in \mathrm{GL}(L_1')$ and extend it by the identity on L_0'. Let the corresponding Levi element of P act on L_1 by $(h_1^{-1})^\vee$ and trivially on L_0 and V_0. A direct check shows that this sends $X_{S_1,T}$ to $X_{S_2,T}$. So the G-orbit is independent of S.

Now assume first that (L_0', B_{T_1}) and (L_0', B_{T_2}) are isometric. Then there exists $h_0 \in \mathrm{GL}(L_0')$ such that

$$B_{T_2}(h_0 u, h_0 v) = B_{T_1}(u, v).$$

Equivalently,

$$T_2 h_0 = (h_0^{-1})^\vee T_1.$$

Extending h_0 by the identity on L_1', and letting the Levi element act on L_0 by $(h_0^{-1})^\vee$, we get

$$X_{S,T_1} \mapsto X_{S,T_2}.$$

Therefore isometric forms give the same G-orbit.

For the converse, separate cases.

If $\epsilon = -1$, then $T_i \in \mathscr{T}^\circ$ is non-degenerate, so Proposition `prop:kernel-image` gives

$$\ker X_{S_i,T_i} = E.$$

Hence any $g \in G$ with

$$g X_{S_1,T_1} g^{-1} = X_{S_2,T_2}$$

must preserve $\ker X_{S_1,T_1} = E$, so $g \in P$. By Proposition `prop:p-normal-form`, two normalized representatives in the same P-orbit have exactly the same residual isometry class. Thus (L_0', B_{T_1}) and (L_0', B_{T_2}) are isometric.

If $\epsilon = +1$ and l is even, then every element of $\mathscr{T}^\circ$ is a non-degenerate alternating form on the even-dimensional space L_0'. Such a form is unique up to isometry, so the converse is automatic.

If $\epsilon = +1$ and l is odd, then every element of $\mathscr{T}^\circ$ is an alternating form of rank $l-1$. Such a form is again unique up to isometry: one chooses a basis

$$e_1, \dots, e_m, f_1, \dots, f_m, r$$

of L_0' in which the form is standard symplectic on the span of the e_i, f_i and vanishes on the radical line Fr. Hence the converse is automatic here as well.

So in all cases,

$$G \cdot X_{S_1,T_1} = G \cdot X_{S_2,T_2} \iff (L_0', B_{T_1}) \cong (L_0', B_{T_2}).$$

proposition prop:multiplicity

statement

Let $x = X_{S,T}$ with $T \in \mathscr{T}^\circ$.

1. If B_T is non-degenerate, then

   $$Z_G(x) \subseteq P.$$

   Hence

   $$m(G \cdot x, P) = 1.$$

2. If $\epsilon = +1$ and l is odd, then B_T has one-dimensional radical

   $$F := \ker T.$$

   In this case

   $$Z_G(x)^\circ \subseteq Z_G(x) \cap P$$

   and

   $$[Z_G(x) : Z_G(x) \cap P] = 2.$$

   Consequently

   $$m(G \cdot x, P) = 2.$$

proof

By Proposition `prop:kernel-image`,

$$\ker x = E \oplus \ker T.$$

If B_T is non-degenerate, then $\ker T = 0$, so $\ker x = E$. Any element of $Z_G(x)$ preserves $\ker x$, hence preserves E, so

$$Z_G(x) \subseteq P.$$

Therefore

$$m(G \cdot x, P) = \frac{|\pi_0(Z_G(x))|}{|\pi_0(Z_G(x) \cap P)|} = 1.$$

Now assume $\epsilon = +1$ and l is odd. Then T has rank $l-1$, so $F = \ker T$ is a line. Also

$$\operatorname{Im} T = \{e \in L_0 : \lambda(e, F) = 0\},$$

so $\operatorname{Im} T$ has codimension 1 in L_0. Choose a line $\ell \subset L_0$ such that

$$L_0 = \operatorname{Im} T \oplus \ell$$

and $\lambda(\ell, F) \neq 0$. More concretely, choose a complement M of F in L_0'; then $T|_M : M \xrightarrow{\sim} \operatorname{Im} T$, and one may take

$$\ell := M^\perp \subset L_0,$$

so that ℓ is a line pairing non-trivially with F and orthogonally to M.
Then

$$\ker x = (L_1 \oplus \operatorname{Im} T) \oplus (\ell \oplus F),$$

and

$$\ker x \cap \operatorname{Im} x = L_1 \oplus \operatorname{Im} T.$$

The quotient

$$\ker x / (\ker x \cap \operatorname{Im} x) \simeq \ell \oplus F$$

is a hyperbolic plane. Its isotropic lines are exactly the two lines ℓ and F. Therefore the r-dimensional totally isotropic subspaces of $\ker x$ containing $\ker x \cap \operatorname{Im} x$ are exactly

$$E = L_1 \oplus \operatorname{Im} T \oplus \ell$$

and

$$E^\sharp = L_1 \oplus \operatorname{Im} T \oplus F.$$

Every element of $Z_G(x)$ preserves both $\ker x$ and $\ker x \cap \operatorname{Im} x$, so it permutes the two-point set

$$\{E, E^\sharp\}.$$

This gives a homomorphism

$$Z_G(x) \to \mathfrak{S}_2.$$

Its kernel is exactly $Z_G(x) \cap P$: if an element lies in the kernel, then it fixes E, so it lies in P; conversely, if an element lies in $Z_G(x) \cap P$, then it preserves E, and since it also preserves the two-point set $\{E, E^\sharp\}$, it must fix both points individually, hence lies in the kernel. Since the target is discrete, the identity component $Z_G(x)^\circ$ lies in the kernel, hence

$$Z_G(x)^\circ \subseteq Z_G(x) \cap P.$$

It remains to show that the permutation is non-trivial. Choose non-zero vectors $f \in \ell$ and $\eta \in F$ with $\lambda(f, \eta) = 1$. Define $\tau \in G$ by:

- τ fixes $L_1 \oplus \operatorname{Im} T \oplus V_0 \oplus L_1'$ pointwise,
- τ fixes the chosen complement M of F in L_0' pointwise,
- $\tau(f) = \eta$ and $\tau(\eta) = f$.

Because $\ell = M^\perp$, the hyperbolic plane $\ell \oplus F$ is orthogonal to $M \oplus \operatorname{Im} T$, and the remaining summands are also orthogonal to it. Hence τ is an isometry. Also x vanishes on both ℓ and F, while x is unchanged on all fixed summands, so $\tau x = x\tau$. Thus

$$\tau \in Z_G(x) \setminus P.$$

Therefore the image of $Z_G(x) \to \mathfrak{S}_2$ is all of $\mathfrak{S}_2$. Because its kernel is $Z_G(x) \cap P$, we get

$$[Z_G(x) : Z_G(x) \cap P] = 2.$$

Since $Z_G(x)^\circ \subseteq Z_G(x) \cap P$, the quotient by component groups equals the ordinary index:

$$m(G \cdot x, P) = \frac{|\pi_0(Z_G(x))|}{|\pi_0(Z_G(x) \cap P)|} = [Z_G(x) : Z_G(x) \cap P] = 2.$$

theorem thm:main

statement

Induced Orbit Representatives and Multiplicity — Self-Contained Problem

Let F be either $\mathbb{R}$ or a non-archimedean local field of characteristic 0. Let $\epsilon \in \{+1, -1\}$.

Setup

Let $(V_0, \langle \cdot, \cdot \rangle_0)$ be a finite-dimensional F-vector space with a non-degenerate bilinear form satisfying

$$\langle v, w \rangle_0 = \epsilon \langle w, v \rangle_0 \qquad \forall v, w \in V_0.$$

Set

$$G_0 := \{g \in \mathrm{GL}(V_0) : \langle gv, gw\rangle_0 = \langle v, w\rangle_0\},$$
$$\mathfrak{g}_0 := \{Y \in \mathrm{End}_F(V_0) : \langle Yv, w\rangle_0 + \langle v, Yw\rangle_0 = 0 \ \forall v, w \in V_0\}.$$

Let $X_0 \in \mathfrak{g}_0$ be nilpotent, and let

$$\mathcal{O}_0 := G_0 \cdot X_0,$$

where G_0 acts on $\mathfrak{g}_0$ by conjugation.
Put

$$c := \dim_F \ker X_0.$$

Fix an integer $r \geq c$ and set $l := r - c$.

Let E, E' be r-dimensional F-vector spaces with a non-degenerate pairing

$$\lambda : E \times E' \to F.$$

Set

$$V := E \oplus V_0 \oplus E'$$

and define a bilinear form on V by

$$\langle e + v + e',\ f + w + f'\rangle := \lambda(e, f') + \epsilon\lambda(f, e') + \langle v, w\rangle_0.$$

Let

$$G := \{g \in \mathrm{GL}(V) : \langle gx, gy\rangle = \langle x, y\rangle\},$$
$$\mathfrak{g} := \{Y \in \mathrm{End}_F(V) : \langle Yx, y\rangle + \langle x, Yy\rangle = 0 \ \forall x, y \in V\}.$$

The group G acts on $\mathfrak{g}$ by conjugation. Let

$$P := \{g \in G : g(E) = E\}.$$

Since $E^\perp = E \oplus V_0$, define

$$\mathfrak{u} := \{Y \in \mathfrak{g} : Y(E) = 0, Y(E^\perp) \subseteq E, Y(V) \subseteq E^\perp\}.$$

Regard $\mathcal{O}_0$ as a subset of $\mathfrak{g}$ by letting each operator in $\mathcal{O}_0$ act on the V_0-summand and act as 0 on $E \oplus E'$.

Define the induced set

$$\mathrm{Ind}_P^G(\mathcal{O}_0) := \overline{G \cdot (\mathcal{O}_0 + \mathfrak{u})} \subseteq \mathfrak{g}.$$

Here the closure is taken in the F-analytic topology on the finite-dimensional F-vector space $\mathfrak{g}$.

A G-orbit $\mathcal{O} \subseteq \mathrm{Ind}_P^G(\mathcal{O}_0)$ is called induced from $\mathcal{O}_0$ via P if it is maximal, among the G-orbits contained in $\mathrm{Ind}_P^G(\mathcal{O}_0)$, for the closure order

$$\mathcal{O}_1 \preceq \mathcal{O}_2 \iff \mathcal{O}_1 \subseteq \overline{\mathcal{O}_2}.$$

For $x \in \mathfrak{g}$, write

$$Z_G(x) := \{g \in G : gxg^{-1} = x\}.$$

For an algebraic subgroup H of G, let H° denote its algebraic identity component, and set

$$\pi_0(H) := H/H^\circ.$$

For $x \in \mathcal{O}_0 + \mathfrak{u}$, define the multiplicity by

$$m(G \cdot x, P) := \frac{|\pi_0(Z_G(x))|}{|\pi_0(Z_G(x) \cap P)|}.$$

In the problem below this is only used for $x = X_{S,T}$.

Construction of the representatives

Choose a subspace $V_+ \subseteq V_0$ such that

$$V_0 = V_+ \oplus \mathrm{Im}\, X_0.$$

Choose decompositions

$$E = L_1 \oplus L_0, \qquad E' = L_1' \oplus L_0',$$

with

$$\dim L_1 = \dim L_1' = c, \qquad \dim L_0 = \dim L_0' = l,$$

such that

$$\lambda(L_i, L_j') = 0 \quad (i \neq j),$$

and such that λ restricts to non-degenerate pairings

$$L_1 \times L_1' \to F, \qquad L_0 \times L_0' \to F.$$

For each isomorphism

$$S : L_1' \xrightarrow{\sim} V_+,$$

define $S^\vee : V_0 \to L_1$ to be the unique linear map satisfying

$$\lambda(S^\vee v, a') = -\langle v, Sa' \rangle_0 \qquad \forall v \in V_0, \forall a' \in L_1'.$$

Define

$$\mathscr{T} := \{T \in \mathrm{Hom}_F(L_0', L_0) : \lambda(Tu, v) + \epsilon\lambda(Tv, u) = 0 \ \forall u, v \in L_0'\}.$$

For $T \in \mathscr{T}$, define a bilinear form on L_0' by

$$B_T(u, v) := \lambda(Tu, v).$$

Then B_T has parity $-\epsilon$, i.e.

$$B_T(v, u) = -\epsilon B_T(u, v).$$

Let $\mathscr{T}^\circ \subseteq \mathscr{T}$ be the subset of maps of maximal possible rank. Equivalently,

$$\operatorname{rank} T = \begin{cases} l, & \epsilon = -1, \\ l, & \epsilon = +1 \text{ and } l \text{ is even}, \\ l-1, & \epsilon = +1 \text{ and } l \text{ is odd}. \end{cases}$$

For an isomorphism $S : L_1' \xrightarrow{\sim} V_+$ and $T \in \mathscr{T}$, define $X_{S,T} : V \to V$ as follows. For

$$e \in E, \qquad v \in V_0, \qquad a' \in L_1', \qquad b' \in L_0',$$

set

$$X_{S,T}(e + v + a' + b') := \left(S^\vee v + Tb'\right) + \left(X_0 v + Sa'\right) + 0 \in E \oplus V_0 \oplus E'.$$

Problem

Prove the following statements.

1. For every isomorphism $S : L_1' \xrightarrow{\sim} V_+$ and every $T \in \mathscr{T}$,

   $$X_{S,T} \in X_0 + \mathfrak{u} \subseteq \mathfrak{g},$$

   where X_0 is regarded as an operator on V by acting on V_0 and acting as 0 on $E \oplus E'$.
2. The induced G-orbits in $\mathrm{Ind}_P^G(\mathcal{O}_0)$ are exactly the orbits

   $$G \cdot X_{S,T} \qquad (T \in \mathscr{T}^\circ),$$

   for any choice of isomorphism $S : L_1' \xrightarrow{\sim} V_+$.
3. The orbit $G \cdot X_{S,T}$ is independent of the choice of S. More precisely, for any two isomorphisms

   $$S_1, S_2 : L_1' \xrightarrow{\sim} V_+$$

   and any $T_1, T_2 \in \mathscr{T}^\circ$,

   $$G \cdot X_{S_1,T_1} = G \cdot X_{S_2,T_2}$$

   if and only if the bilinear spaces

   $$(L_0', B_{T_1}) \qquad \text{and} \qquad (L_0', B_{T_2})$$

are isometric, meaning that there exists $h \in \mathrm{GL}(L_0')$ such that

$$B_{T_2}(hu, hv) = B_{T_1}(u, v) \qquad \forall u, v \in L_0'.$$

4. For $S : L_1' \xrightarrow{\sim} V_+$ and $T \in \mathscr{T}^\circ$, compute

$$m(G \cdot X_{S,T}, P) = \frac{|\pi_0(Z_G(X_{S,T}))|}{|\pi_0(Z_G(X_{S,T}) \cap P)|}.$$

As part of the computation, prove that

$$Z_G(X_{S,T})^\circ \subseteq Z_G(X_{S,T}) \cap P,$$

and hence that

$$m(G \cdot X_{S,T}, P) = [Z_G(X_{S,T}) : Z_G(X_{S,T}) \cap P],$$

where the right-hand side is the ordinary index of the corresponding groups of F-points. The required answer is

$$m(G \cdot X_{S,T}, P) = \begin{cases} 1, & B_T \text{ is non-degenerate}, \\ 2, & \epsilon = +1 \text{ and } l \text{ is odd}. \end{cases}$$

proof

Assertion (1) is Lemma `lem:parametrize-x0-plus-u`.

Assertion (2) is Proposition `prop:induced-orbits`.

Assertion (3) is Proposition `prop:s-independence-and-orbit-criterion`.

Assertion (4) is Proposition `prop:multiplicity`.

This proves all four requested statements.